

ControlTower AI

An enterprise engineering operations command center that turns GitHub activity and operational signals into release decisions, blocker triage, action briefs, and delivery-ready communications.

Release Readiness

Merge Analytics

Explainable Scoring

Gemini-Enhanced Summaries

Slack & Telegram Delivery

4

Primary panels: release risk, incident hotspots, blocked work, and action brief.

3

Operational outcomes: decision support, escalation drafting, and audit-ready summaries.

2

Decision layers: deterministic heuristics for trust plus Gemini for executive language.

Problem

- Release readiness is still judged manually across PR queues, issue backlog, and scattered tools.
- Merge throughput and blocker pressure are hard to convert into a clear go/no-go decision.
- Operational updates and escalation messages are repeatedly written by hand.

Solution

- Ingest live GitHub repo signals into one operations-ready dataset.
- Score release risk, merge pressure, and blocker queues with explainable heuristics.
- Generate shareable outputs for Slack and Telegram from the same run.

Current Product Highlights

- Live GitHub sync with open PR, merged PR, open issue, and closed issue totals.
- Merge analytics: 7d / 30d merged PRs and average review-to-merge time.
- Top 10 ranked views for merged PRs, oldest open PRs, and issue backlog.
- Release decision banner with high-risk slice detection and owner guidance.
- Selectable shareable content: repo intelligence, action brief, risk detail, release brief.
- Slack and Telegram delivery for selected operational content.

S**Situation**

Engineering teams face growing release complexity, rising review latency, and fragmented visibility across GitHub and operations workflows.

T**Task**

Build a practical intelligent enterprise solution that turns engineering signals into trustworthy release guidance and communication-ready outputs.

A**Action**

Built a Next.js command center with GitHub-derived operational slices, deterministic scoring, Gemini refinement, merge analytics, ranked repo views, and direct delivery to Slack and Telegram.

R**Result**

ControlTower AI compresses manual release triage into one workflow: sync a repo, understand risk, review evidence, generate an action brief, and share the right message immediately.

Why It Matters

- Moves from dashboarding to operational decision support.
- Creates trust with explainable heuristics before applying AI language generation.
- Bridges analysis and action with direct, shareable outputs.

Next Step

- Expand from GitHub into Slack, Jira, PagerDuty, and deploy systems.
- Add historical benchmarks for merge velocity and blocker recurrence.
- Strengthen enterprise auth for private repo analysis.

Developer

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Closing Statement

ControlTower AI demonstrates how enterprise AI can move beyond passive analytics and into daily engineering operations: faster release decisions, better visibility into merge and backlog pressure, and immediate distribution of the right message to the right channel.